

	Course Syllabus EE 493 Power Electronics SPRING 2025-26
---	---

INSTRUCTOR	
Name	<i>Umut Başaran</i>
E-mail	<i>umut.basaran@ozyegin.edu.tr</i>
Phone Number	-
Office Number	<i>AB4 – B416</i>
Office Hours	<i>by appointment</i>
Research/Teaching Assistants	-

COURSE ORGANIZATION	
Interaction Methods	<i>Lectures will be face-to-face in class. LMS will be used for announcements.</i>
Technology Used by the Instructor	<i>Laptop, LTSpice (or similar), MATLAB/Simulink</i>
Technology Requirements for Students	<i>Laptop, LTSpice (or similar), MATLAB/Simulink. Students need a computer that should be able to run circuit simulation programs. LTSpice is necessary for circuit simulations.</i>

COURSE MATERIALS	
Required Books	-
Recommended Books	<i>Issa Batarseh, & Ahmad Harb. (2018). Power Electronics: Circuit Analysis and Design. Springer.</i> <i>Muhammad H. Rashid, Power Electronics Handbook, 4th Edition (2018), Butterworth, Heinemann, Imprint of Elsevier</i>
Other Materials	-
Accessing Course Materials	<i>Any other lecture related materials will be uploaded to LMS</i>

WEEKLY SUBJECTS (TENTATIVE)		
Week	Date	Subject
1	10.02.2026	Active Components used in Power Electronics
2	17.02.2026	Active Components used in Power Electronics and Power Loss
3	24.02.2026	Passive Components used in Power Electronics & Thermal Analysis of Active Components
4	03.03.2026	Thermal Analysis of Active Components
5	10.03.2026	LDOs and DCDC Converters – Part 1
6	17.03.2026	LDOs and DCDC Converters – Part 2
7	24.03.2026	Bootstrap and Charge Pump Circuits – Part 1
8	31.03.2026	Midterm Exam - Bootstrap and Charge Pump Circuits – Part 2
9	07.04.2026	DC AC Converters (Inverters) – Part 1
10	14.04.2026 (Withdrawal Week)	DC AC Converters (Inverters) – Part 2
11	21.04.2026	AC DC Converters
12	28.04.2026	Power Factor Correction (PFC) and PFC Circuits
13	05.05.2026	Electromagnetic Compatibility (EMC) Analysis of Power Electronic Circuits – Part 1
14	12.05.2026	Electromagnetic Compatibility (EMC) Analysis of Power Electronic Circuits – Part 2

ASSESSMENT METHODS, WEIGHTS AND RULES			
Type	Weight	Implementation Rule	Makeup Rule

Final Exam	35%	The final exam will be open books and notes. No communication devices will be allowed during the exam.	BÜT exam will serve as the makeup for the final exam.
Midterm Exam	35%	The midterm exam will be open books and notes. No communication devices will be allowed during the exam.	Only valid excuses with an official report are accepted to qualify for a midterm makeup. At most one makeup will be given in the course due to health reports.
Project	30%	After the midterm exam, a project will be assigned. The project will involve the design and simulation of a power electronics circuit in LTspice (or similar), and students will be required to submit a technical report.	There will be no makeup for the project.
Total	100%		

METHODS FOR TRACKING ATTENDANCE & IMPACT OF ATTENDANCE ON SUCCESS

Method	Application Details	Required Tools and Make-up Method	Impact of Attendance of Course Grade
QR Code (MyOzU)	During class, a QR code will be displayed on the screen. Scanning this code with the MyOzU app on your mobile device will be accepted as proof of attendance. You can access the guideline here	You must have a smartphone with the MyOzU app installed. If you do not have such a device, inform the instructor during class and sign the attendance sheet.	Participation in less than 3 lectures will automatically result in an F or U grade. In-class digital attendance will be held each week at 11:00. Out of 14 sessions, at least 3 must be attended. Each additional attendance beyond 3 adds 0.5 bonus points (out of 100) to the final grade.

ASSESSMENT(S) BEFORE THE WITHDRAWAL DEADLINE

Type of Assessment(s)	Weight(s)	Date by When the Score of the Assessment is to be Announced
Midterm Exam	35%	10.04.2026

DETERMINING LETTER GRADE

- A bell curve will be used to determine the letter grade.
- Students should receive 40 or more points from the final exam to pass the course.

EXPECTED STUDENT SEMESTER WORKLOAD

Activities Carried Out in The Presence of An Instructor	Total Hours in Semester
Lecture (Including Discussion Sessions)	42
Recitation/Practice	20
Laboratory/Studio/Kitchen/Flight	-
Field Work	-
Activities Carried Out by The Learner Him/Herself	Total Hours in Semester
Working on Assignments/Project	50
Pre-class Learning of Course Material	30
Review of Course Material for Exam Preparation	30
Other	-
Overall Total Hours in Semester	162

RULES FOR THE USE OF AI IN THE COURSE

Use of AI in the Course (Free Use within Ethical Framework)	You may freely use AI tools (ChatGPT, Copilot, Gemini, Claude, Grok, DeepSeek, etc.) in this course, provided that you do so ethically and in compliance with the Principles for the Responsible Use of AI listed below. You may benefit from AI in assignments and other learning activities. However, you must clearly acknowledge AI contributions, act ethically and transparently, and comply with university policies .
Principles for the Responsible Use of AI	1. Use AI as a tool to support your learning, not as the ultimate source of knowledge: AI can assist your learning and work, but the final judgment, creativity, and ownership of the output must always remain with you. 2. Be transparent about AI's contribution: If you use AI to generate, summarize, or improve any part of your work, you must clearly acknowledge it. Transparency protects your academic honesty and strengthens professional integrity. 3. Recognize the limitations and biases of AI: AI may produce inaccurate, incomplete, or biased outputs. Always review the results critically and verify them before using. 4. Consult diverse sources, not just AI: Do not rely solely on AI for information. Support, compare, and cross-check AI outputs with other valid and reliable sources. 5. Protect your data and privacy: Never share personal, sensitive, or confidential information with AI tools. Safeguarding such information is your responsibility.

OTHER RULES AND INFORMATION	
Health	The mode of instruction is subject to change in case of pandemic or other similar unavoidable reasons in line with the directives of the authorized public bodies. In such a case, the necessary revisions will be binding for all members of the OzU community, and will be announced by the relevant units, in particular Student Services.
Safety	Announced safety procedures must be strictly followed during any laboratory and similar work. Please refer to the safety guidelines posted in the laboratories. In case of emergencies, call 9911 (216 564 9911).
Accessibility	Reasonable accommodations will be provided for students with verifiable disabilities. Please inform your instructor if you need any assistance.
Academic and Technical Support	Students can get help from the Writing Center, Solution Center and IT in case they need academic and/or technical support with their classes.
Objections to Examination Results	Students who object to their exam results and cannot resolve their objections with the instructor of the course, may file a written petition for a review of their exam papers at their Dean's/Director's Office within five business days from the announcement their exam results.
Academic Integrity and Honor Code	Students are assumed to abide by scholastic honesty and uphold the Honor Code. Scholastic dishonesty includes, but is not limited to cheating, attempting to cheat, plagiarizing, fabricating information or citations, facilitating acts of dishonesty by others, having unauthorized possession of examinations, submitting the work of another person, or tampering with the academic work of other students. Any form of scholastic dishonesty is a serious academic violation and will result in a disciplinary action and also may result in student getting a 0 grade in the related assessment.
Unauthorized Recording/Sharing of Lectures	Recording/sharing of synchronous/asynchronous lectures or copying of lecture recordings are prohibited without the permission of the instructor. Actions such as unauthorized recording, giving the recording to someone else, taking it, using it or sharing it in physical or virtual media are prohibited and/or are offenses according to the relevant legislation. In such cases, legal procedures will be initiated in addition to starting disciplinary actions against those involved.
Flexibility	Circumstances may arise during the course that prevents the instructor from fulfilling each and every component of this syllabus; therefore, the syllabus is subject to change. Students will be notified prior to any changes.
Principles for Classroom Behavior	Creating a safe space where ideas can be freely discussed in our classroom is everyone's responsibility. In accordance with the relevant disciplinary regulations, any verbal or written expression that disrupts the order of the class, involves the unauthorized distribution of materials, or is discriminatory or violent, and that violates the dignity of individuals in the classroom (students or educators) is strictly prohibited. During class, all participants are expected to respect each other's rights.